

Acoustics Exam One

Report

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Abstract:

This examination aims to test our abilities to collect data in an organized error minimization fashion. This also includes testing our knowledge on high and low pass filter circuits as well as our ability to critically think/ analyze our work.

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I. Introduction:

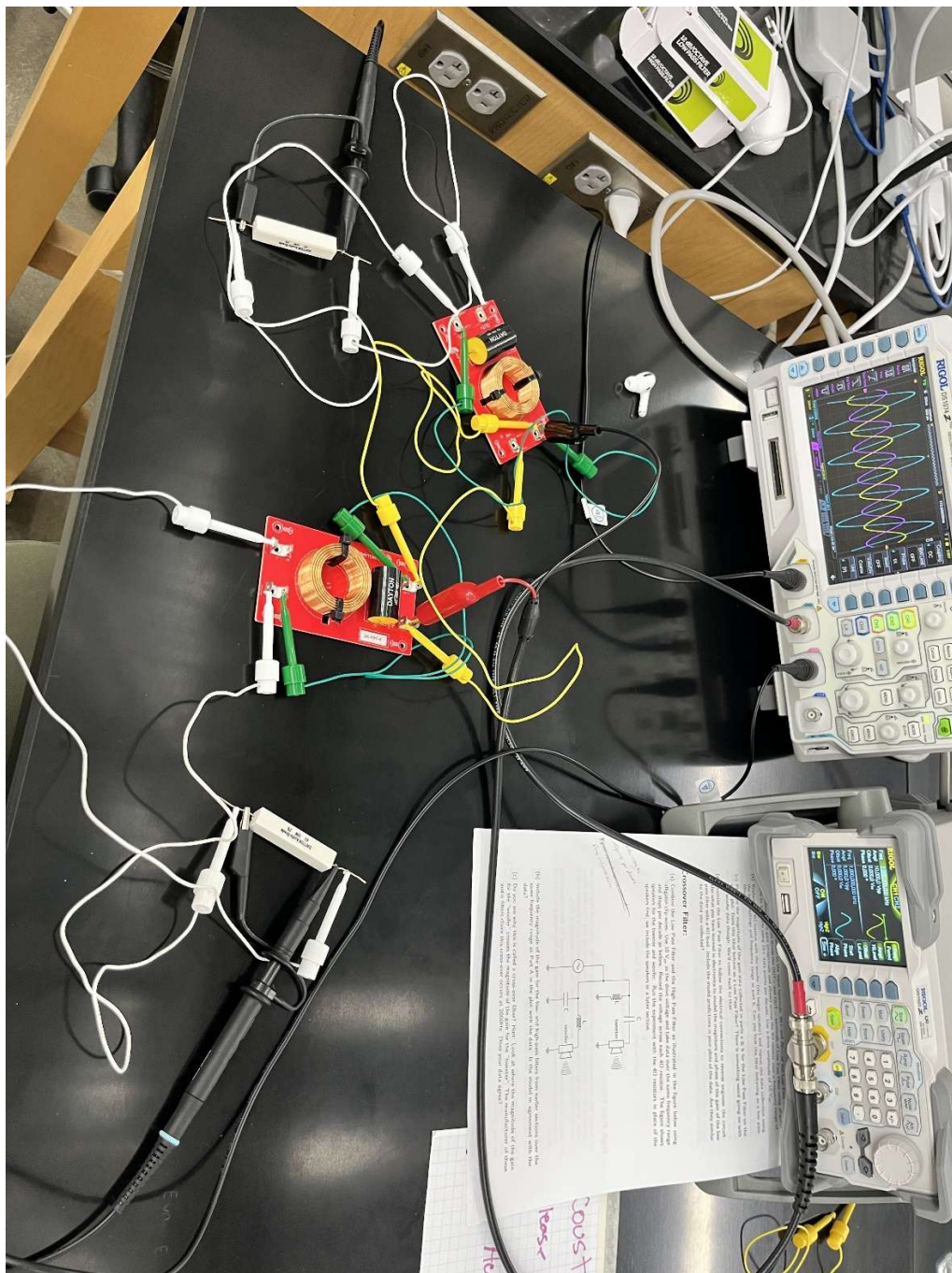
Acoustics and electronics have become very close to each other ever since we figured out how to use electronics to produce quality sound. In acoustics analog filters like the high and low pass filters are essential in this bond between the two studies. The math involved in the electronic and acoustic physics behind this exam are very similar and play off one another.

Audio signals contain frequencies ranging from low to high and in between. To produce sound properly the higher frequencies must be passed to a tweeter speaker while the low frequencies must be passed to a woofer speaker. In terms of acoustics and just physics in general, the tweeter is ideal for the high frequencies due to its ability to rapidly oscillate its diaphragm. The case is the same for the woofer, but since low frequencies out of a speaker move less air if they were the same size as the tweeter, the speaker must be larger to move air properly.

Equipment List & Setup:

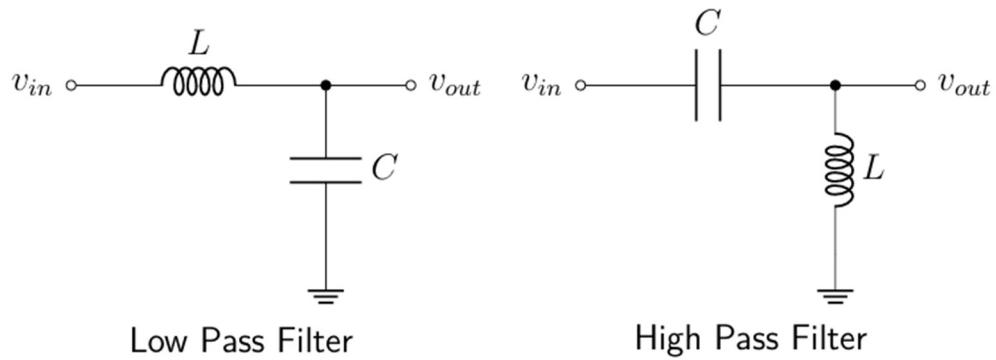
ITEM	NEEDED
High Pass Filter	1
Low Pass Filter	1
4 Ohm Resistor	2
4 Ohm Tweeter	1
4 Ohm Woofer	1
Oscilloscope	1
Signal Generator	1
Connecting wires	assortment

Preview image of the Cross Over Resistors setup:

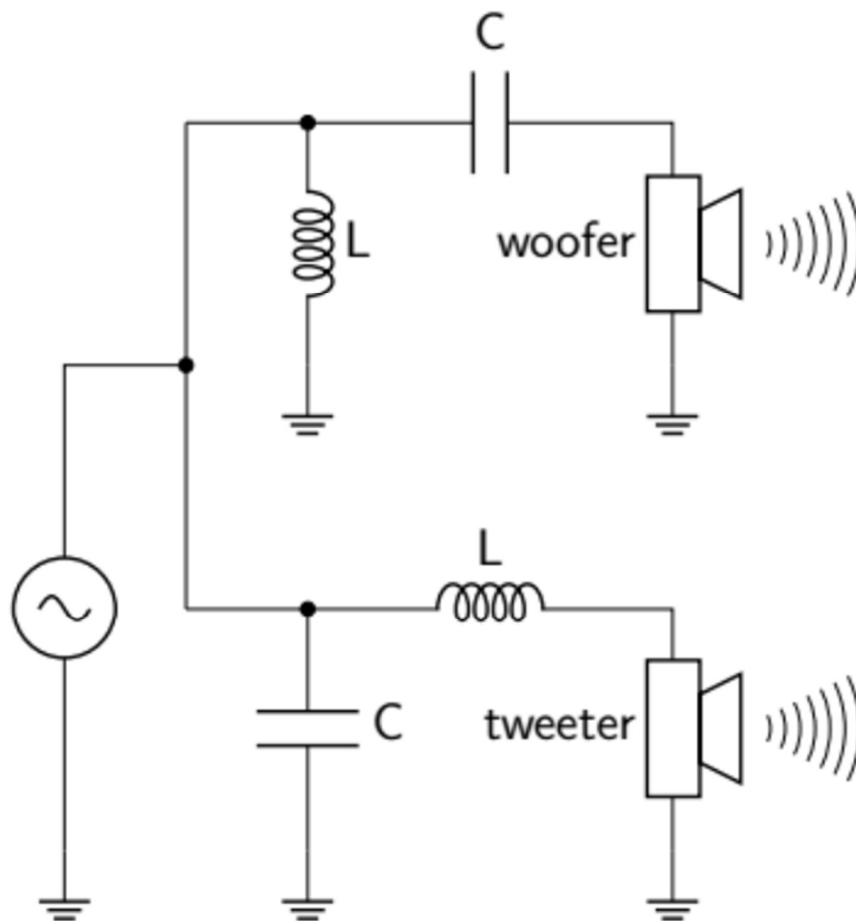


II. Theory:

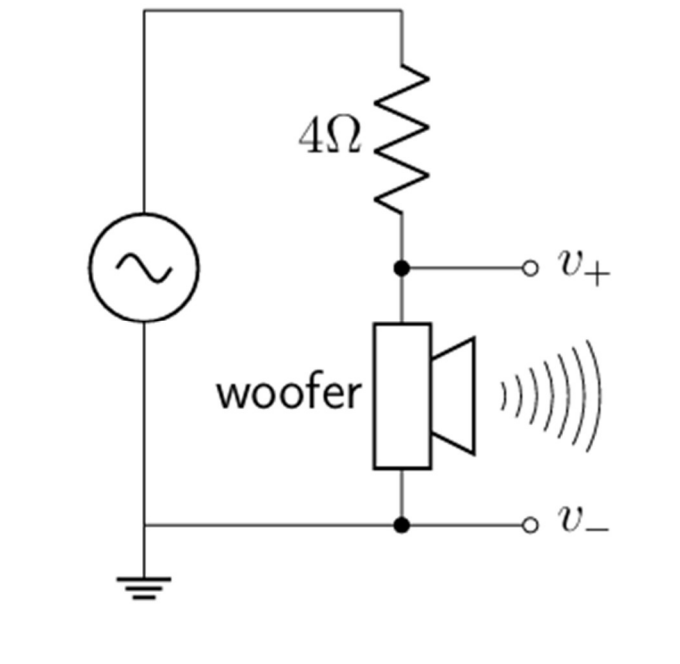
A. Circuits:



Cross over Circuit:



Driver Frequency Circuit:



B. Background:

RLC circuits have a complex impedance defined by:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

What this tells us is that the impedance or just resistance of the RLC circuit changes as the frequency changes. This is what will play a huge role in explaining the differences between the plots for the circuits with resistors vs. with the speakers.

In the high-pass filter we take the output voltage across the resistor and divide it by the input voltage (which also changes with the impedance) to get the gain. At high frequencies, the inductor's impedance is high while the capacitor's is low. This allows for current to pass to the load (resistor or speaker).

Inversely, in low-pass filters at low frequencies the inductor's impedance is low while the capacitor's is high. This means that the inductor allows current to pass while the capacitor blocks it. This then delivers power to the load. The gain function is the same for both.

In the beginning of the exam description we are told that the cross over point happens at about 2000 Hz. This is the claim of the manufacturer. With the measured values of the inductor and capacitor we can theoretically estimate the crossover frequency using this equation below:

$$f_c = \frac{1}{2\pi\sqrt{LC}} = 1989 \text{ Hz}$$

So in terms of theoretical calculations the manufacturer is correct. However, we will soon see that this is for the ideal attached load and not the speakers.

III. Experimental Procedure:

Measurements will be made on each circuit (high pass, low pass, cross over, and driver frequency). The circuit diagrams were given above in the theory sections if you need visual context.

For the high pass filter, we want to measure the gain (magnitude and phase) across the 4 Ohm resistor over a frequency range of 10 Hz to 10,000 Hz. Ideally you want to collect at least 10 data points per decade. I would recommend starting at 10 and increasing the frequency by 5 Hz per data point until you get to 100 Hz. After this increase the frequency by 50 Hz until you reach 1000 Hz. Finally increase the frequency by 500 Hz until you reach 10,000 Hz. Once this is done replace the 4 Ohm resistor with the 4 Ohm tweeter and collect more data using the same method. Use 10 V peak to peak as the input amplitude from the function generator.

The same detailed process will be carried out when collecting data using the low pass filter. This time replace the 4 Ohm resistor with the woofer.

In taking data for the crossover filter, it is a good idea to collect more data points around the 2000 Hz mark to verify the manufacturers' words. You will first use a 4 Ohm resistor connected in series with the circuits then replace the 4 Ohm resistor with the tweeter on the high pass and vice versa on the low pass. Collect data in the same manner as the previous two circuits. However, the manufacturer also claims that the filters have a 12 dB/Octave roll off. The roll off is the straight

sloping part of the gain's magnitude. In audio engineering the octave is used instead of a decade. An octave is a doubling in frequency. Hence, going from 100 Hz to 200 Hz is one octave; going from 2000 Hz to 8000 Hz would be 2-octaves, etc. As you're taking data with the 4 Ohm resistors (ideal load no complex impedance), ask yourself if the theoretical calculations and the experimental data match the manufacturer's claims.

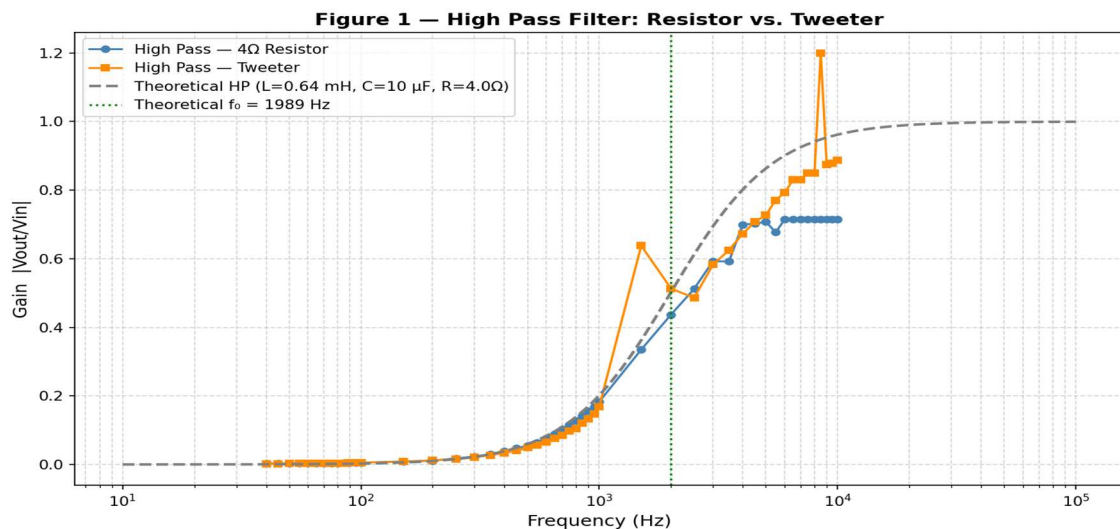
For the driver frequency response set up the circuit as seen in the theory section. First with the 4 Ohm resistor and woofer in series. Second with the 4 Ohm resistor and tweeter in series. Lastly place both resistors in series. Measure the voltage out across the woofer, tweeter, and resistor. The same frequency range and number per decade should be used. We want to use this data to plot the magnitude of the gain over these frequencies. What do you notice about the woofer and tweeters response compared to just the resistor?

IV. Results:

Note: To make my life easier, I made sure to collect data in Excel spreadsheets for each data collection. This was then converted to .csv files for easy data calling when making my plots. The results below will display a table and then a plot of my results. Some notes might be thrown in.

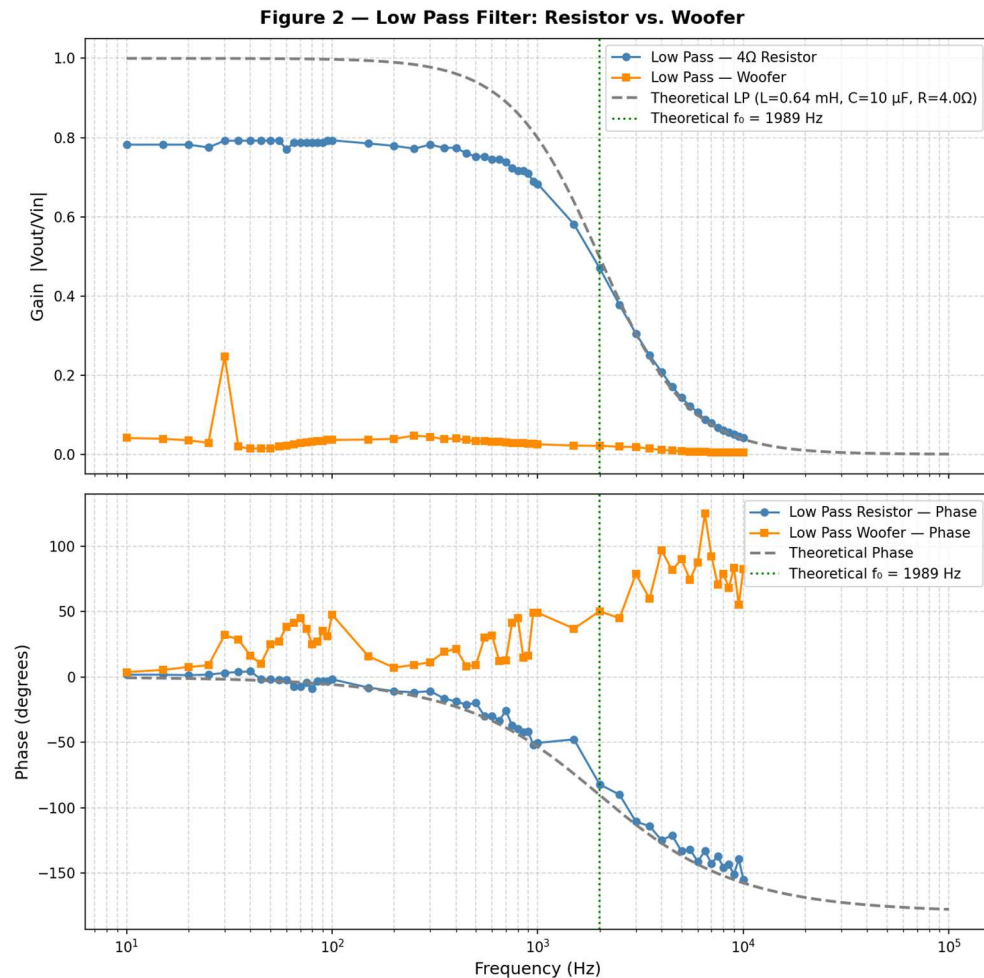
A. The High-Pass Circuit:

Observation	Expected	Actual
Gain at 40 Hz	~0 (blocked)	0.003
Gain at 10 kHz	~0.7 – 0.8 (passband)	0.715
Tweeter Gain at 1.5 kHz	Should peak near resonance	0.638 – Resonance peak visible



B. The Low-Pass Circuit:

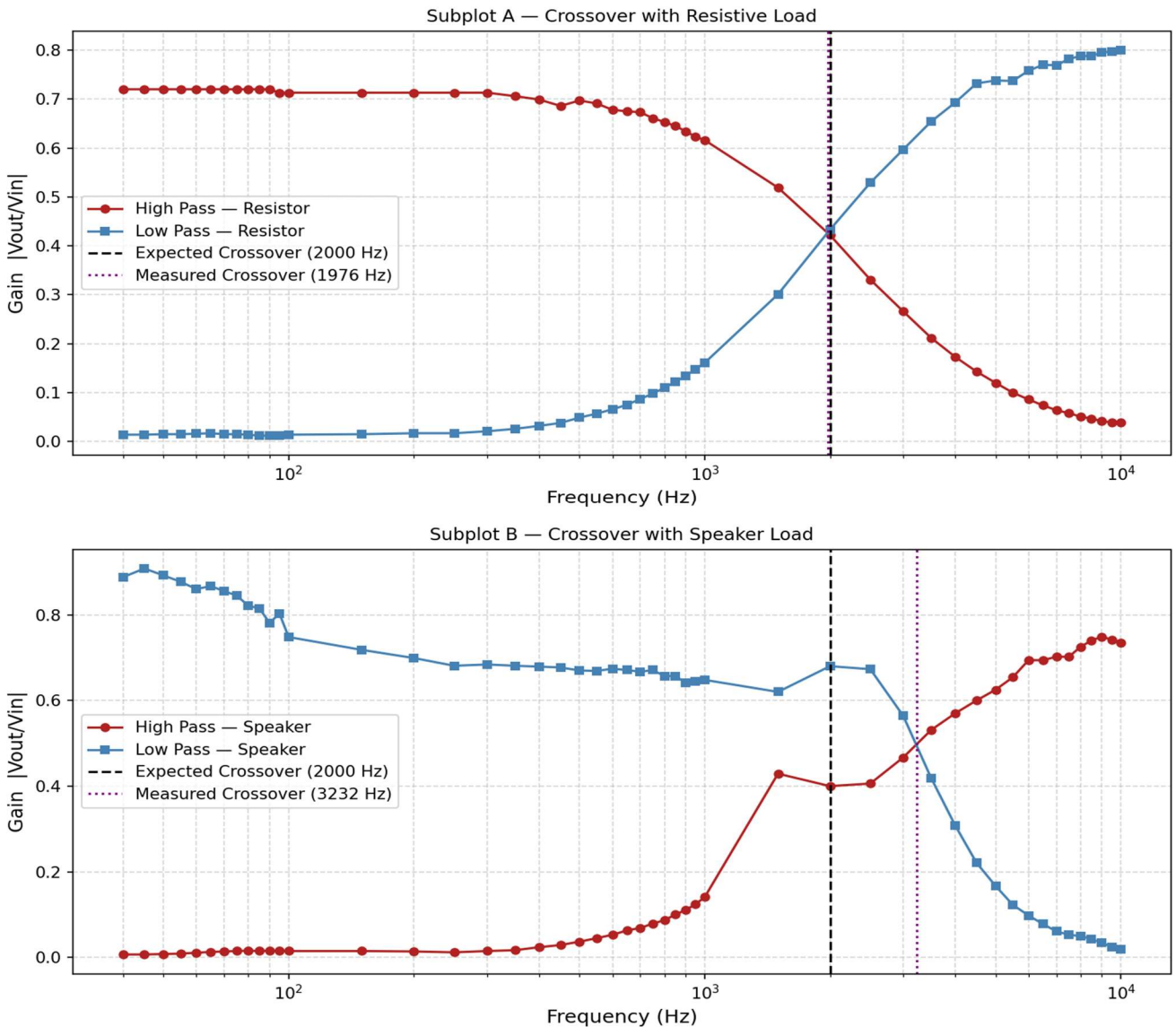
Observation	Expected	Actual
Gain at low frequency	~0.78 (passband)	0.782
Gain at 10 kHz	~0 (blocked)	0.041
Woofer gain low frequency	Should be low (mechanical)	0.041



C. The Crossover Circuit:

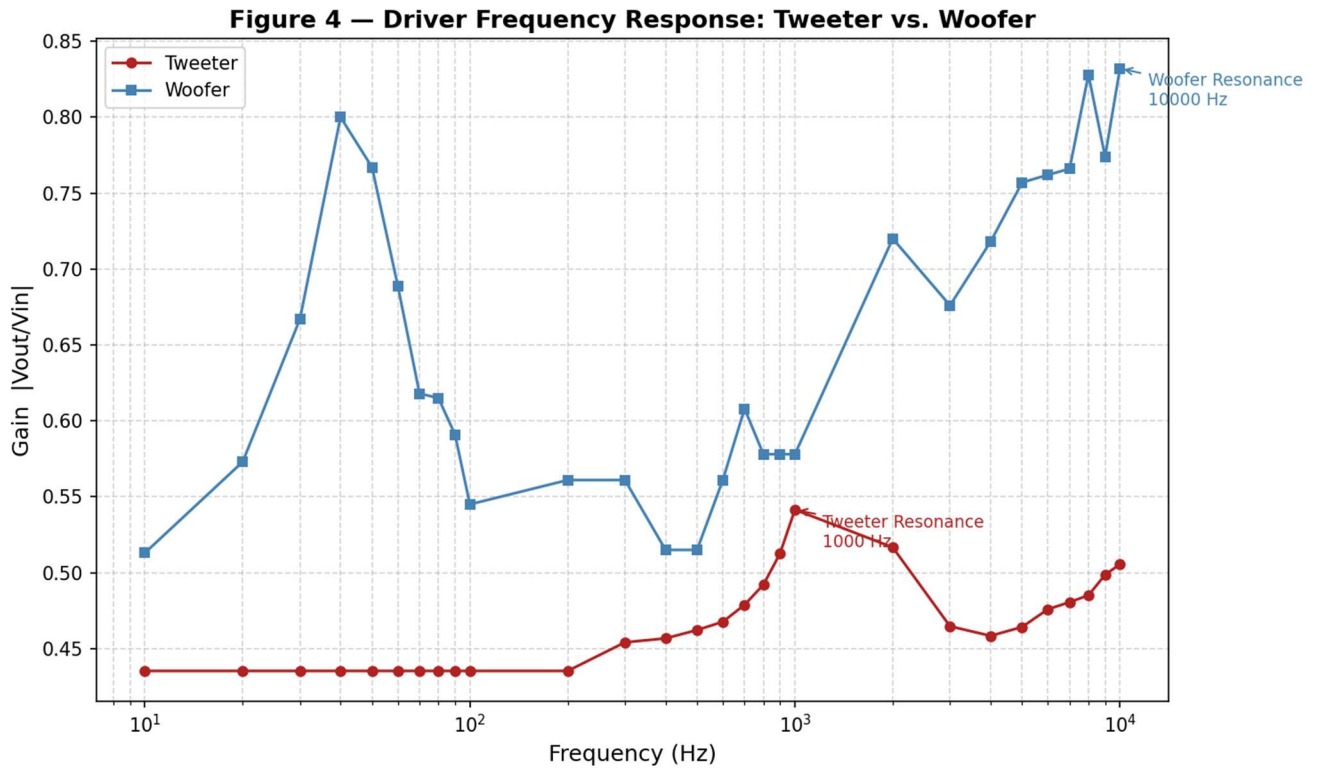
Observation	Expected	Actual
HP gain at 40 Hz	~0	0.014
LP gain at 40 Hz	~0.72	0.720
Crossover (~2000 Hz)	HP $\approx$ LP	HP=0.422, LP=0.433

Figure 3 — Crossover Filters: Resistors & Speakers



D. Driver Frequency Response:

Driver	Expected	Actual
Woofer	Peak at low frequency resonance	Peak ~40 Hz (0.800)
Tweeter	Flat at low frequency, rises	Flat until ~900 Hz, peaks at 1000 Hz (0.541)



V. Discussion:

A. High-Pass Filter:

The high-pass filter data collection and plots are the best I got. The plots fit the expected trajectories of the data. There were a couple of spikes in the plots, but this is due to human error when taking measurements. The tweeter resonance is around 1500 Hz based on the peak gain of 0.638 V. This makes total sense for a small tweeter driver.

B. Low-Pass Filter:

Now the lowpass filter data had unexpected results in terms of the gain over frequency plot. Its totally flat! However, all my numbers point to that being correct. This is likely a problem with my code or in my data. I am fairly satisfied with the results I got I just need to work on my plotting skills. The cross over filter Is the fun one to talk about.

C. Crossover Filter:

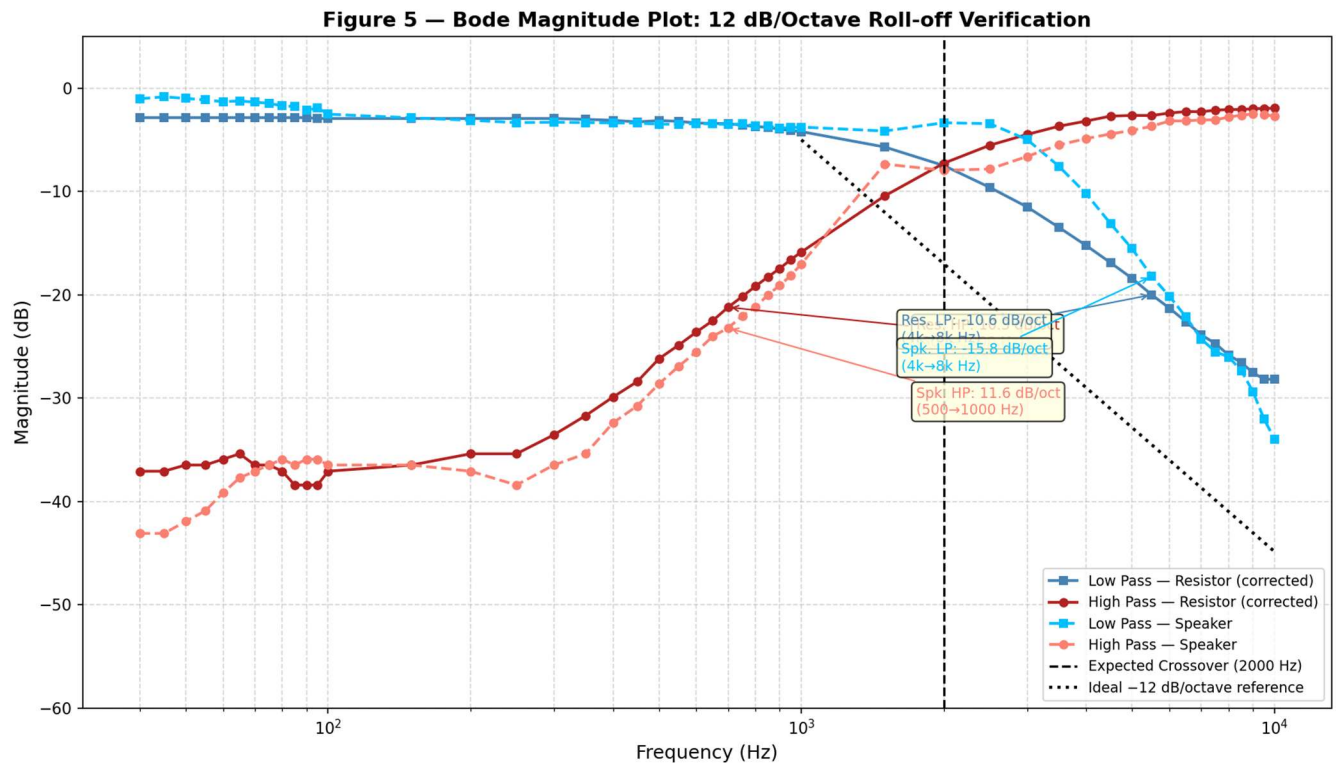
These crossover filter plots are some of the most beautiful plots I've ever made. From the data to the tightening of the lines in the plot to that perfect crossover point which highlights exactly what we expected. In the resistor crossover image you see the low pass and high pass basically mirror each other with a clear and defined cross over frequency. This was experimentally calculated to be cross over at 1976 Hz! Wonderful! But wait why would the speaker cross over results be different?

To answer the above question, we need to go back to the theory section and remind ourselves of complex impedances. Now, it is easy to get resistance and impedance confused with each other. In a way they are both just resistances, but its not that simple. The way I think of it is impedance is active resistance that changes as a response to other things happening. The reason this is important is because recall the design of a speaker. It's a coil of copper wire around or in a ferromagnetic ring or cylinder (depending on design). That coil of copper wire that provides the AC current to bounce the speaker diaphragm up and down acts as an inductor!

What happens as a result is the load is no longer purely resistive. It is now a complex impedance that changes based on frequency. At higher frequencies the inductor like speakers become very inductive. This causes a shift in the cross over point. They still meet in a good point at the same time because they are in series and have similar designs.

Now the question of the 12 dB/ Octave claim still remains. Does the crossover filter roll off at the manufacturer's claimed 12 dB/Octave rate? Well this is what plots and tables are for. After plotting this I found the information in the table below as well as the plot below:

```
=====
Resistor HP roll-off (500→1000 Hz): 10.3 dB/octave
Speaker HP roll-off (500→1000 Hz): 11.6 dB/octave
Resistor LP roll-off (4k→8k Hz): -10.6 dB/octave
Speaker LP roll-off (4k→8k Hz): -15.8 dB/octave
Expected: ±12 dB/octave (2nd order)
=====
```



This is great and all but what exactly does this tell us? The real speaker impedance is actually not exactly 4 Ohms. Also the woofer inductance modifies the response.

D. Driver Frequency Response:

The main thing that I want to answer for this one is the question of does the driver frequency response show the mechanical resonance?

In-fact it does! Look back at the graph in the results section for this circuit and experiment. The massive spikes in gain at specific frequencies are the mechanical resonance frequencies. At these frequencies the physical speaker cone needs very little electrical energy to vibrate with great amplitude.

VI. Conclusions:

To conclude this report I'd like to remind myself and whomever reads this to make sure you set things up right before you collect data. Its like a physics problem where half of the problems importance is setting it up right and visualizing it and the rest is just the math. I found errors in my .csv files where the lowpass and high pass data were flipped. I also found that I accidentally calculated the gain as input voltage over output voltage instead of the other way around. There was also a typo in the instructions regarding the cross over circuit and when my data didn't seem right I took a second look at our set up. I found out that we connected the woofer to the highpass filter and the tweeter to the lowpass. Of course, we had to collect data again which was a pain, but the plots and experience was worth it.

Overall, this is a fun lab to do especially if you like learning about audio circuits. Every modern car with a decent sound system has some form of cross over circuit in it to guide high frequency signals to the tweeter and low frequency signals to the woofer. I always found myself turning down the mids in my equalizer in my truck which would be around the area of crossing over. Perhaps I just found out why I instinctually do that!

VII. References:

Note: I need help with proper referencing

- [1] “Exam_1_2026”, Dr. Slaton Ph.D., 2026
- [2] CoPilot in VSCode, Creating plots and calculations.
- [3] [Series RLC Circuit and RLC Series Circuit Analysis](#)